

January 30, 2026

Continuing 1.5:

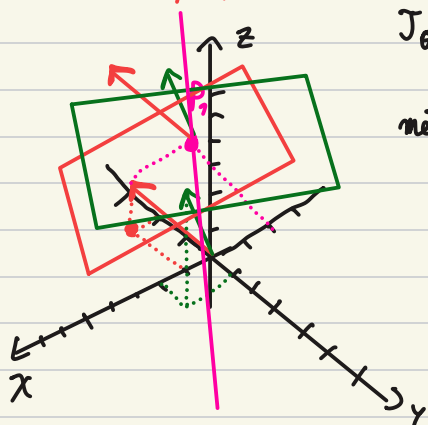
e.g. (section 1.5 example 5) Line on which two non-parallel planes intersect

$$\pi_1: x - 2y + z = 4$$

$$\vec{n}_1 = (1, -2, 1)$$

$$\pi_2: 2x + y + 3z = -7$$

$$\vec{n}_2 = (2, 1, 3)$$



To define a line we need a point and a direction (or two distinct points)

method 1: 1) Find a point P_1 of intersection:

$$\text{What if } z=0: \pi_1 \rightarrow x - 2y = 4$$

$$\pi_2 \rightarrow 2x + y = -7$$

$$P_1(-2, -3, 0)$$

$$5x = -10 \quad x = -2 \quad y = -3$$

$$(-2) - 2(-3) + (0) = 4 \quad 2(-2) + (-3) + 3(0) = -7$$

2) Find a direction vector $\vec{v}$ for the line:

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 1 \\ 2 & 1 & 3 \end{vmatrix} = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & 3 \end{vmatrix} \begin{matrix} \hat{i} \\ \hat{j} \\ \hat{k} \end{matrix}$$

$$= (-6-1)\hat{i} + (2-3)\hat{j} + (1-4)\hat{k}$$

$$= -7\hat{i} - \hat{j} + 5\hat{k}$$

$$\vec{r}(t) = \begin{pmatrix} -2 \\ -3 \\ 0 \end{pmatrix} + t \begin{pmatrix} -7 \\ -1 \\ 5 \end{pmatrix}$$

method 2: 1) Find two points P_1, P_2 of intersection:

$P_1(-2, -3, 0)$ is known, now let's say $x=0$:

$$-2y + z = 4, \quad y + 3z = -7$$

$$P_2(0, -\frac{19}{7}, -\frac{10}{7})$$

$$7z = 4 - 14 = -10 \quad z = -10/7$$

$$y = -19/7$$

2) Compute displacement vector $\vec{P_1P_2}$:

$$\vec{v} = \vec{P_1P_2} = (0, -\frac{19}{7}, -\frac{10}{7}) - (-2, -3, 0)$$

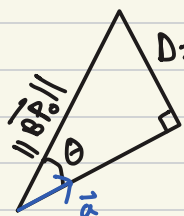
3) line equation $\vec{r} = \vec{P_1} + t\vec{v}$:

(section 1.5 example 7) Distance between a point P_0 and a line l :



Method 1 (works in any $\mathbb{R}^n$):
distance = $\| \vec{BP}_0 - \text{proj}_{\vec{a}} \vec{BP}_0 \|$

Method 2 (only in $\mathbb{R}^3$)



$$D = \text{distance} = \| \vec{BP}_0 \| \sin \theta$$

$$\text{recall } \| \vec{u} \times \vec{v} \| = \| \vec{u} \| \| \vec{v} \| \sin \theta \quad (\text{in } \mathbb{R}^3)$$

$$\text{so } D = \frac{\| \vec{a} \times \vec{BP}_0 \|}{\| \vec{a} \|}$$

eg: Distance between point $P_0(2,1,3)$ and line $\vec{l}(t) = t \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}$

$$\vec{a} = (-1, 1, -2)$$

$$\| \vec{a} \| = \frac{\sqrt{1^2 + 1^2 + 2^2}}{\sqrt{6}}$$

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B point in the line? $B(2,3,-2)$

$$\begin{aligned} \vec{BP}_0 &= (2,1,3) - (2,3,-2) \\ &= (0, -2, 5) \end{aligned}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & -2 \\ 0 & -2 & 5 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} \\ -1 & 1 \end{vmatrix} \hat{k} + \begin{vmatrix} \hat{i} & \hat{k} \\ -1 & -2 \end{vmatrix} \hat{j} + \begin{vmatrix} \hat{j} & \hat{k} \\ 1 & -2 \end{vmatrix} \hat{i} = (5 - 4)\hat{i} + (0 - (-5))\hat{j} + (2 - 0)\hat{k} = (1, 5, 2)$$

$$\| \vec{a} \times \vec{BP}_0 \| = \sqrt{1^2 + 5^2 + 2^2} = \sqrt{30}$$

$$D = \frac{\sqrt{30}}{\sqrt{6}} = \sqrt{5}$$

(section 1.5 example 8) Distance between planes inside $\mathbb{R}^3$
 If the two planes are not parallel, then they intersect somewhere
 (distance is 0)

Distance between parallel planes inside $\mathbb{R}^3$

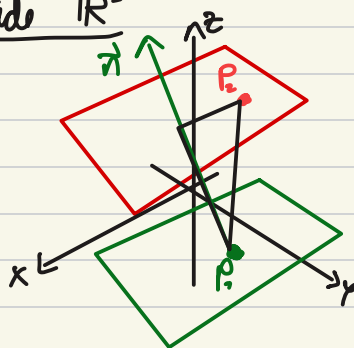
1) Find a normal vector $\vec{n}$

2) Find a point P_1 in Π_1 and P_2 in Π_2

3) Distance $D = \|\text{proj}_{\vec{n}} \overrightarrow{P_1 P_2}\|$

e.g. $\Pi_1: 2x - 2y + z = 5$

$\Pi_2: 2x - 2y + z = 20$



1) $\vec{n} = (2, -2, 1)$

2) $P_1(1, 1, 5)$ $P_2(1, 1, 20)$

3) $\overrightarrow{P_1 P_2} = (1, 1, 20) - (1, 1, 5) = (0, 0, 15)$

$$\text{proj}_{\vec{n}} \overrightarrow{P_1 P_2} = \frac{\vec{n} \cdot \overrightarrow{P_1 P_2}}{\vec{n} \cdot \vec{n}} \vec{n} = \frac{15}{9} (2, -2, 1)$$

$$\vec{n} \cdot \vec{n} = 4 + 4 + 1 = 9$$

$$\vec{n} \cdot \overrightarrow{P_1 P_2} = (2)(0) + (-2)(0) + (15)(1) = 15$$

$$\begin{aligned} \left\| \frac{15}{9} (2, -2, 1) \right\| &= \left| \frac{15}{9} \right| \|(2, -2, 1)\| = \frac{15}{9} \sqrt{4 + 4 + 1} \\ &= \frac{15}{9} 3 = \frac{15}{3} = 5 \end{aligned}$$